Roadmap. Srednicki §§14–15: the one-loop self-energy of φ^3 in $d = 6$. LecNote11's pole at the physical m with residue 1 fixes A, B via $\Pi(-m^2) = \Pi'(-m^2) = 0$; dim reg gives $\Pi = \frac{1}{2}\alpha \int dx\, D \ln(D/D_0) - \frac{\alpha}{12}(k^2 + m^2)$, and $\operatorname{Im}\Pi$ gives $\rho(s)$. Next: Z_g (LecNote13, §16), $2 \to 2$ (LecNote14, §20), MS-bar and the RG (LecNote15, §§27–28). Supplements: FeynmanRulesLSZ, iPiassignment, DimRegMSbar, SpectralRepresentation, DimensionalAnalysis_hbar, ThreeRegulators.

Loop corrections to the propagator (Sec 14)

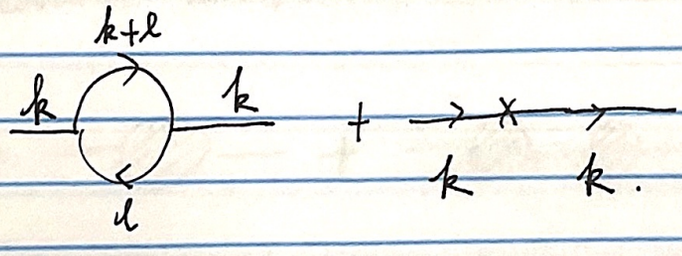
In path integral, we wrote the exact propagator as

$$\frac{1}{i}\Delta(x_1 - x_2) \equiv \langle 0|T\,\varphi(x_1)\varphi(x_2)|0\rangle = \delta_1\delta_2 iW(J)\Big|_{J=0}$$

$iW(J)$ is the sum of all connected diagrams

leading order (tree level) ——— which is $\Delta(x_1 - x_2)$

next-to leading order



using Feynman Rules

$$\frac{1}{i}\hat{\Delta}(k^2) = \frac{1}{i}\Delta(k^2) + \frac{1}{i}\hat{\Delta}(k^2)\left[i\Pi(k^2)\right]\frac{1}{i}\hat{\Delta}(k^2) + O(g^4)$$

where $\hat{\Delta}(k^2) = \dfrac{1}{k^2 + m^2 - i\epsilon}$

which is the free-field propagator.

from the diagram we can read that

$$i\Pi(k^2) = \frac{1}{2}(-ig)^2 \left(\frac{1}{i}\right)^2 \int \frac{d^4l}{(2\pi)^4} \tilde{\Delta}((l+k)^2)\,\tilde{\Delta}(l^2)$$

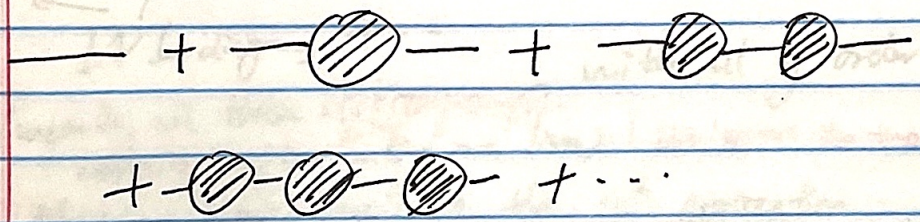
($Z_g = 1 + O(g^2)$)

symmetry factor

$$= -i\,(Ak^2 + Bm^2) + O(g^4)$$

$A \leftarrow (Z_\phi - 1)$, $B \leftarrow (Z_m - 1)$.

but we can see these effort can be further summed via geometric series.



$$\frac{1}{i}\tilde{\Delta}(k^2) = \frac{1}{i}\tilde{\Delta}(k^2) + \frac{1}{i}\tilde{\Delta}(k^2)\left[i\Pi(k^2)\right]\frac{1}{i}\tilde{\Delta}(k^2)$$

$$+ \frac{1}{i}\tilde{\Delta}(k^2)\left[i\Pi(k^2)\right]\frac{1}{i}\tilde{\Delta}(k^2)\left[i\Pi(k^2)\right]\frac{1}{i}\tilde{\Delta}(k^2)$$

$$+ \cdots\cdots$$

$$\tilde{\Delta}(k^2) = i\tilde{\Delta}(k^2)\,\frac{1}{1 - \Pi(k^2)\tilde{\Delta}(k^2)}$$

$$= \frac{1}{k^2 + m^2 - i\epsilon - \Pi(k^2)}$$

1 Why "iff": near $-m^2$, $\tilde{\Delta} \approx 1/[(1-\Pi'(-m^2))(k^2+m^2)-\Pi(-m^2)]$; Π is smooth there (its cut starts at $-4m^2$). Residue 1 is LSZ's $\langle k|\varphi(0)|0\rangle = 1$; the canonically normalized field is $Z_\varphi^{1/2}\varphi$, with residue $Z_\varphi \le 1$ (LecNote11). With both conditions $\Pi = O((k^2+m^2)^2)$, so p. 2's series converges near the pole.

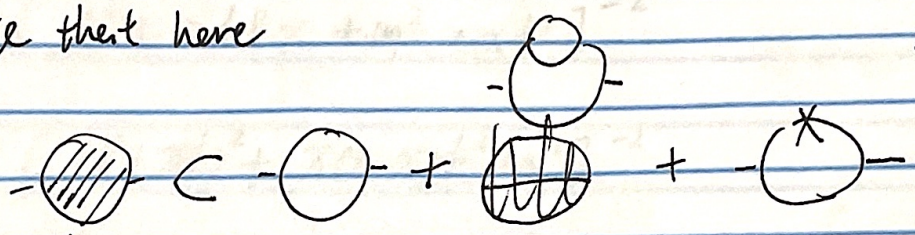
We know that the exact propagator has a pole at $k^2 = -m^2$ with residue of one, this means (only and only if)

$$\Pi(-m^2) = 0$$

$$\Pi'(-m^2) = 0 \quad \left(\left.\frac{d\Pi(k^2)}{dk^2}\right|_{k^2=-m^2} = 0 \right) \text{①}$$

This is equivalent to canonical quantization $\langle k|k'\rangle = \dots$

Note that here



one-particle irreducible ← 1PI diagrams + ...

(meaning all those still connected after cutting one line) with all g^2 order corrections (any cuts). [We ignore the two external propagators]

Now we can evaluate the g^2 correction

(let's do the integral)

[This avoids double counting of the geometric series we did earlier]

To evaluate, we need to invoke

"Feynman's formula" that. (to combine denominators)

$$\frac{1}{A_1 A_2 \cdots A_n} = \int dF_n \, (x_1 A_1 + x_2 A_2 + \cdots + x_n A_n)^{-n}$$

Here the integration is over Feynman parameter x_i

$$\int dF_n = (n-1)! \int_0^1 dx_1\, dx_2\, dx_3 \cdots dx_n\, \delta(x_1+x_2+x_3+\cdots+x_n-1).$$

Let's prove such an identity in HW.

Hence, knowing one more trick, we now can calculate.

$$\Delta(k+l)^2\,\Delta(l^2) = \frac{1}{(l^2+m^2)\left((k+l)^2+m^2\right)}$$

$$= \int_0^1 dx \left[x\left((l+k)^2+m^2\right) + (1-x)(l^2+m^2)\right]^{-2}$$

$$= \int_0^1 dx \left[l^2 + 2kl\,x + m^2 + xk^2\right]^{-2}$$

$$= \int_0^1 dx \left[(l+xk)^2 + x(1-x)k^2 + m^2\right]^{-2}$$

$$= \int_0^1 dx \left[q^2 + D\right]^{-2}$$

$$q \equiv l + xk, \qquad D \equiv x(1-x)k^2 + m^2.$$

The jacobian for changing variable from $d^4l \longrightarrow d^4q$ is trivially one.

Now as q^0 and q^i have different signs in the metric. let's think about q^0. as ~~q^2~~ q appears in squares (why not? the I. I. quantity)

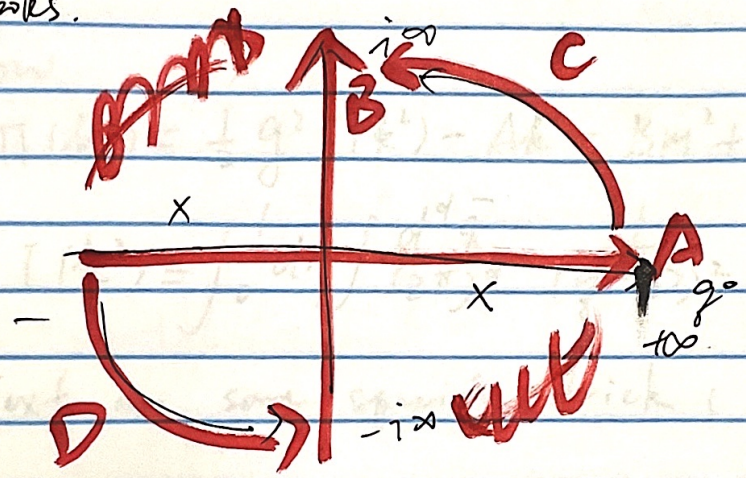
Let's perform the **Wick Rotation**

$$q^0 \longrightarrow i q^0.$$

If the integrand vanishes fast enough as $|q^0| \to \infty$

this integral won't change value with such rotation
(contour)

Here, it are super helpful in keeping track of the poles.



$A + C - B \to 0$

$C, D \to 0$

So $A = B$

With such rotation, we change the integration to Euclidean d-dimensional $\bar{q}$, with

$q^0 = i\bar{q}_d$ and $q_j = \bar{q}_j$

$q^2 = \bar{q}^2$

with $\bar{q}^2 = \bar{q}_1^2 + \bar{q}_2^2 + \dots + \bar{q}_d^2$

$d^d q = i\, d^d \bar{q}$

So $\int_{-\infty}^{\infty} dq^0 f(q^0)$

$= \int_{-\infty}^{+\infty} d\tilde{q}^0 f(i\tilde{q}^0)$ [1]

Wick Rotation avoid pole to make the calculation simple

[1] Should read $\int dq^0 f(q^0) = i \int d\tilde{q}^0 f(i\tilde{q}^0)$: on the imaginary axis $dq^0 = i\, d\tilde{q}^0$, the same i as $d^d q = i\, d^d \bar{q}$ in the text.

Then

$$\int d^d q\, f(q^2 - i\epsilon) = i \int d^d \bar{q}\, f(\bar{q}^2).$$

as long as $f(\bar{q}^2) \to 0$ faster than ~~$1/q^d$~~ $1/\bar{q}^d$ as $\bar{q} \to \infty$.

Now.

$$\Pi(k^2) = \frac{1}{2} g^2 I(k^2) - A k^2 - B m^2 + O(g^4).$$

$$I(k^2) \equiv \int_0^1 dx \int \frac{d^d \bar{q}}{(2\pi)^d} \frac{1}{(\bar{q} + D)^2}.$$

Next are some specific trick ( ).[1]

$$\Pi''(k^2) = \frac{1}{2} g^2 I''(k^2) + O(g^4)$$

(this gets rid of A & B terms) (for our expression for φ^3 theory)

$$I''(k^2) = \int_0^1 dx\, 6x^2(1-x)^2 \int \frac{d^d \bar{q}}{(2\pi)^d} \frac{1}{(\bar{q} + D)^4}.$$

Skip

We can perform this integral first and then $\int k^2$ twice (which allows us to match b.c.

$$\Pi(-m^2) = 0, \quad \Pi'(-m^2) = 0.$$)

[1] Optional, but it is the whole calculation without a regulator: at $d = 6$, $I'' = (4\pi)^{-3} \int dx\, x^2(1-x)^2/D$ is finite, so $\Pi'' = \frac{1}{2}\alpha \int dx\, x^2(1-x)^2/D$. Integrating twice with $\Pi(-m^2) = \Pi'(-m^2) = 0$ gives p. 13's Π exactly. Divergences sit only in $a + bk^2$, which A, B absorb. p. 8 uses the $d < 8$ bound.

This integral $\int d^d q \frac{1}{(q^2+D)^4}$

is convergent up to $d<8$

~~this~~ while the original $I(k^2)$ integration up to $d<4$.

This new range contains $d=6$.

Skip

Let's view it again from Taylor expansion around $k^2=-m^2$ point of view.

$$\Pi(k^2) = i\left[\frac{1}{2}g^2 I(-m^2) + (A-B)m^2\right] \quad \cdots \quad d\geq 4$$

$$+\left[\frac{1}{2}g^2 I'(-m^2) - A\right](k^2+m^2). \quad \cdots \quad d\geq 6$$

$$+\frac{1}{2!}\frac{1}{2}g^2 I''(-m^2)\Big](k^2+m^2)^2 + \cdots \quad \cdots \quad d\geq 8$$

$$+\bar{U}(g^4).$$

It's clear that $I^{(n)}(-m^2)$ is divergent for $d\geq 4+2n$.

~~so~~ but A can cancel the divergence at 2nd line from I'

and $A-B$ can cancel the divergence of 1st line from I.

no more free parameters and $\Pi(k^2)$ now is finite for $d<8$.

For $d>8$, the procedure breaks down.

the theory is non-renormalizable.[1]

(and we will see this through dim-analysis)

1 Should read: for $d \geq 8$ the one-loop Π needs a third (k^4) counterterm. Non-renormalizability starts at $d > 6$: $[g] = (6-d)/2 < 0$ (LecNote14, §18; DimensionalAnalysis_hbar); in $d = 7$ two loops already need a φ^4 counterterm.

Now it is about how to handle (keep track of) infinities.

Method A. (mostly commonly used in earlier textbooks but also useful for clarifying many physics confusions) (we can have a brief discussion about it in the RGE part).

Pauli-Villars regularization.
(adding an effective cut-off scale to the theory)

$$\Delta(p^2) \longrightarrow \frac{1}{p^2+m^2-i\epsilon} \frac{\Lambda^2}{p^2+\Lambda^2-i\epsilon}$$

Λ is the ultraviolate cutoff, this will make $d<8$ finite as it would be ~~$\int d^d q \, \Delta(q^2) \delta I$~~

$$\int d^d q \frac{1}{(q^2+1)^4}$$

such a term also ad hoc violates gauge symmetry so one needs to be careful.

Still, we can calculate and then take the $\Lambda \to \infty$ limit, that makes the $\alpha\Lambda^2, \alpha\Lambda^4$ parts cancel. [1]

Λ tracks the divergencies.

1 Should read: A, B fixed by $\Pi(-m^2) = \Pi'(-m^2) = 0$ cancel the divergent part, $\frac{1}{2}\alpha[\frac{1}{2}\Lambda^2 - (\frac{k^2}{6} + m^2)\ln\Lambda^2]$ in $d = 6$ (no Λ^4); then $\Lambda \to \infty$ gives p. 13's Π, with $\ln\Lambda^2 \leftrightarrow 2/\epsilon$.

Method B: Dimensional Regularization
(Dim - Reg).

Very popular & beautiful, (t'Hooft & Veltman).

One can directly evaluate from a dimension & analytically continue.

the angular part of $d\bar{q}$ over whole ~~area~~ range yields the area (angular volume) of a unit sphere.

$$\Omega_d = \frac{2\pi^{d/2}}{\Gamma(\frac{1}{2}d)}.$$

(one can verify $\int d^d\bar{q}\, e^{-\bar{q}^2}$) $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt.$

$\Gamma(x)$ is the Euler Gamma function.

$$\Gamma(n+1) = n!$$

$$\Gamma(n+1/2) = \frac{(2n)!}{n!\, 2^{(2n)}} \sqrt{\pi}$$

$$\Gamma(-n+\varepsilon) = \frac{(-1)^n}{n!}\left[\frac{1}{\varepsilon} - \gamma + \sum_{k=1}^{n} k^{-1} + \mathcal{O}(\varepsilon)\right]$$

$\gamma = 0.5772\ldots$

Euler Mascheroni Constant.

$$\gamma = \lim_{n\to\infty}\left(-\log n + \sum_{k=1}^{n}\frac{1}{k}\right)$$

$$= \int_1^\infty \left(-\frac{1}{x} + \frac{1}{\lfloor x \rfloor}\right) dx, \quad \lfloor\cdot\rfloor \text{ is the floor function.}$$

The radial part is

$$\int \frac{d^d\bar{q}}{(2\pi)^d} \frac{(\bar{q}^2)^a}{(\bar{q}^2+D)^b} = \frac{\Gamma(b-a-\frac{1}{2}d)\Gamma(a+\frac{1}{2}d)}{(4\pi)^{d/2}\Gamma(b)\Gamma(\frac{1}{2}d)} D^{-(b-a-d/2)}$$

Very useful formula (Eq. 14.27).

For the theory we are evaluating, $a=0$, $b=2$

(General case) For general d, the coupling g has a mass dimension $\epsilon/2$

$$\epsilon \equiv 6-d.$$

To account for this, we introduce a new parameter $\tilde{\mu}$.

$$g \to g\tilde{\mu}^{\epsilon/2}$$

~ not a physical parameter.

(hence g remains dimensionless for general theory)

$\tilde{\mu}$ is unphysical so no observable can depend on it

(later we revisit and derived RGE).

Setting $d = 6-\epsilon$, we get

$$I(k^2) = \frac{\Gamma(-1+\epsilon/2)}{(4\pi)^3} \int_0^1 dx\, Q \left(\frac{4\pi}{D}\right)^{\epsilon/2}.$$

With the substitution of $g \to g\tilde{\mu}^{\epsilon/2}$ [we'll see this convenience later]

and defining $\alpha \equiv \frac{g^2}{(4\pi)^3}$ (not the fine-structure constant, which is $\frac{e^2}{4\pi}$)

In normal THEP counting it would be $\alpha \equiv \frac{g^2}{4\pi}$, and order-by-order $\frac{\alpha}{(4\pi)^2}$

$$\Pi(k^2) = \frac{1}{2}\alpha\,\Gamma\left(-1+\frac{\epsilon}{2}\right)\int_0^1 dx\, D\left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\epsilon/2} - Ak^2 - B\mu^2 + O(\alpha^2)$$

D: dimless.

Taking the $\epsilon \to 0$ limit.

$$y^{\epsilon/2} = 1 + \frac{\epsilon}{2}\ln y + O(\epsilon^2)$$

and $\Gamma(-1+x)$

$$\Pi(k^2) = -\frac{1}{2}\alpha\left[\left(\frac{2}{\epsilon}+1\right)\left(\frac{1}{6}k^2+\mu^2\right) + \int_0^1 dx\, D\ln\left(\frac{4\pi\tilde{\mu}^2}{e^{\gamma}D}\right)\right] - Ak^2 - B\mu^2 + O(\alpha^2)$$

(noting that $D = x(1-x)k^2+\mu^2$

$$\int_0^1 dx\, D = \frac{1}{6}k^2+\mu^2).$$

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It is convenient to define

$$\mu \equiv \sqrt{4\pi}\, e^{-\gamma/2}\tilde{\mu}$$

So we have now

$$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D/m^2)$$

$$- \left\{ \frac{1}{6}\alpha \left[\frac{1}{\epsilon} + \ln(\mu/m) + \frac{1}{2} \right] + A \right\} k^2$$

$$- \left\{ \alpha \left[\frac{1}{\epsilon} + \ln(\mu/m) + \frac{1}{2} \right] + B \right\} m^2 + O(\alpha^2)$$

If we demand

$$A = -\frac{1}{6}\alpha \left[\frac{1}{\epsilon} + \ln(\mu/m) + \frac{1}{2} + \kappa_A \right] + O(\alpha^2)$$

$$B = -\alpha \left[\frac{1}{\epsilon} + \ln(\mu/m) + 1/2 + \kappa_B \right] + O(\alpha^2)$$

$$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D/m^2)$$

$$+ \alpha \left(\frac{1}{6}\kappa_A k^2 + \kappa_B m^2 \right) + O(\alpha).$$

now $\Pi(k^2)$ is finite! and independent of μ.

To fix κ_A and κ_B, we must still impose the condition.

$\Pi(-m^2) = 0$ & $\Pi'(-m^2) = 0$ [1]

1 Imposing them (p. 13): $\kappa_A = \frac{7}{3} - \frac{\pi\sqrt{3}}{2} \approx -0.387$, $\kappa_B = \frac{11}{12} - \frac{\pi\sqrt{3}}{6} \approx 0.0098$; two linear conditions fix two constants uniquely (p. 14's question), and $\Pi(0) = \alpha\kappa_B m^2$. This is the on-shell scheme (m = pole mass). MS-bar (LecNote15, §27) keeps only $A = -\alpha/6\epsilon$, $B = -\alpha/\epsilon$.

Noting that

Eq 14.41 → $\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D) + \text{linear in } k^2 \text{ and } m^2 + O(\alpha^2)$

we can impose

$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D/D_0) + \text{linear in } (k^2+m^2) + O(\alpha^2)$

here $D_0 \equiv D\big|_{k^2=-m^2} = [1-x(1-x)]\, m^2$

We can then differentiate Eq 14.41 w.r.t. k^2

and find ~~$\Pi'(-m^2)$~~ $\Pi'(-m^2)$ vanishes for

$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D/D_0) - \frac{1}{12}\alpha (k^2+m^2) + O(\alpha^2)$

↑ this integral can be done "normally"

and there is a "branch point" at $k^2 = -4m^2$ [1]

the exact propagator can now be written as

$$\tilde{\Delta}(k^2) = \left(\frac{1}{1-\Pi(k^2)/(k^2+m^2)}\right)\frac{1}{k^2+m^2-i\epsilon}$$

1 Where the branch point comes from: keep the $-i\epsilon$ dropped on p. 4, $D \to D - i\epsilon$. For $k^2 = -s < -4m^2$, $D < 0$ on $\frac{1}{2}(1-\beta) < x < \frac{1}{2}(1+\beta)$, $\beta = (1-4m^2/s)^{1/2}$, and $\ln(D - i\epsilon) = \ln|D| - i\pi$: $\operatorname{Im}\Pi(-s) = \frac{\pi\alpha}{12} s\beta^3 > 0$ from LecNote11's threshold $s = 4m^2$. Then (p. 15) $\rho = \frac{\alpha}{12} s\beta^3/(s-m^2)^2 \geq 0$ (SpectralRepresentation).

Let's see the logic here,
we are essentially try to find solutions that matches the boundary conditions.
(Q for yourself: is the assignment unique?)

In any case, we found

$$\Pi(k^2) = \frac{1}{2}\frac{g^2}{(4\pi)^3}\int_0^1 dx\, D \ln(D/D_0) - \frac{1}{12}\frac{g^2}{(4\pi)^3}(k^2+m^2) + O(g^4)$$

↓ this can be done in a closed form.

$$\alpha \equiv \frac{g^2}{(4\pi)^3}$$

$$\Pi(k^2) = \frac{1}{12}\alpha\left[C_1 k^2 + C_2 m^2 + 2k^2 f(\gamma)\right] + O(\alpha^2)$$

$$f(\gamma) = \gamma^3 \tanh^{-1}(1/\gamma), \quad C_1 = 3 - \pi\sqrt{3}$$

$$\gamma = (1 + 4m^2/k^2)^{1/2}, \quad C_2 = 3 - 2\pi\sqrt{3}.$$

↓ there is a square root branch point at $k^2 < -4m^2$

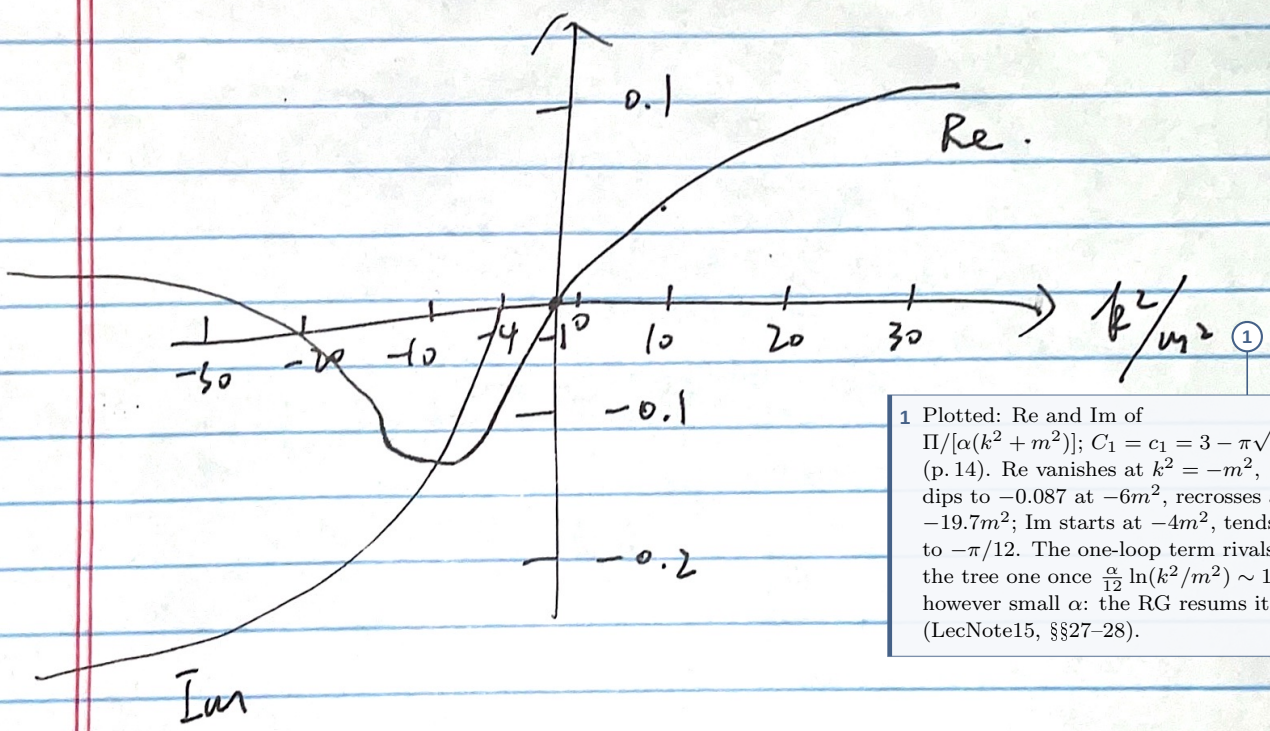
Recall that

$$\tilde{\Delta}(k^2) = \tilde{\Delta}(k^2)\frac{1}{1 - \Pi(k^2)\tilde{\Delta}(k^2)}$$

For large values of $|k^2|$

$$\frac{\Pi(k^2)}{k^2+m^2} \simeq \frac{1}{12}\alpha\left[C_1 + \ln\frac{k^2}{m^2}\right]$$

k^2 being negative & $\ln(|k^2| - i\epsilon) = \ln|k^2| - i\pi$



1 Plotted: Re and Im of $\Pi/[\alpha(k^2+m^2)]$; $C_1 = c_1 = 3 - \pi\sqrt{3}$ (p. 14). Re vanishes at $k^2 = -m^2$, dips to -0.087 at $-6m^2$, recrosses at $-19.7m^2$; Im starts at $-4m^2$, tends to $-\pi/12$. The one-loop term rivals the tree one once $\frac{\alpha}{12}\ln(k^2/m^2) \sim 1$, however small α: the RG resums it (LecNote15, §§27–28).

Without much extra work, one can make connection to the Lehmann–Källén form.

Seeks $\underbrace{\pi\rho(s)}_{\frac{1}{\pi}\mathrm{Im}\,\Pi} = \dfrac{\mathrm{Im}\,\Pi(-s)}{(-s+m^2-\mathrm{Re}\,\Pi(-s))^2 + (\mathrm{Im}\,\Pi(-s))^2}$.

spectral density, which starts to grow and ~~negat~~ $k^2 \lesssim -4m^2$